



Provably Convergent Image Regularization using Learning-based 1-Lipschitz Denoisers

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Inverse problems

$$\mathbf{y} = \mathcal{H}(\mathbf{x}_g, \boldsymbol{\eta}),$$

- $\mathbf{x}_g \in \mathbb{R}^n$ - Ground truth data (image or video).
- $\mathbf{y} \in \mathbb{R}^m$ - Measurements.
- $\boldsymbol{\eta} \in \mathbb{R}^m$ - Random Noise.
- $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ - Degradation Model.

Inverse Problem: Estimate $\mathbf{x}_g$ given $\mathbf{y}$.

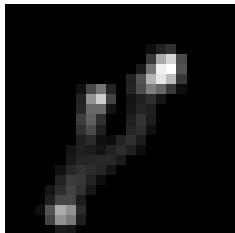
Reconstruction applications - Examples

$$\mathbf{y} = \mathbf{B}\mathbf{x}_g + \boldsymbol{\eta},$$

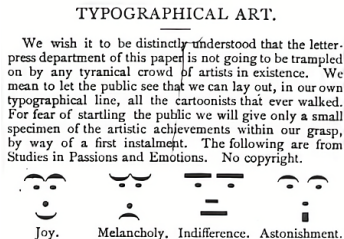
- $\mathbf{B} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ - Linear blur kernel.
- $\boldsymbol{\eta} \in \mathbb{R}^n$ - Gaussian Noise.



(a) Blurred image.



(b) Blur kernel.



(c) Expected output.

Deblurring

Reconstruction applications - Examples

$$\mathbf{y} = \mathbf{S}\mathbf{x}_g + \boldsymbol{\eta},$$

- $\mathbf{S} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ - Sensing matrix (identity matrix with missing rows).
- $\boldsymbol{\eta} \in \mathbb{R}^n$ - Gaussian Noise.



(a) Measurements.



(b) Sensing mask.



(c) Expected output.

Inpainting

Reconstruction applications - Examples

Imaging modality

- Image super-resolution
- Denoising
- Poisson Image Denoising
- X-Ray Computed Tomography
- MRI reconstruction
- Compressed Sensing
- Synthetic aperture radar imaging
- Hyperspectral-Multispectral image fusion

Video modality

- Video stabilization
- Video superresolution
- Video frame interpolation
- Flicker removal
- Video deblurring
- Video inpainting/ object removal
- Color restoration
- 3D reconstruction from videos

Inverse problems are ill-posed

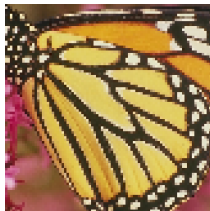
- A unique solution may not exist.
- The solution may not depend continuously on the measurement.



(a) y .



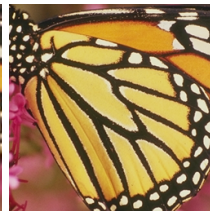
(b) $\mathbf{S}^T y$.



(c) Nearest interpolation.



(d) Bicubic interpolation.



(e) Expected output.

Superresolution: $\mathbf{y} = \mathbf{S}\mathbf{x}$, $\mathbf{S}$ is sampling matrix.

Solvers for inverse problems

- Bayesian Approach
- End-to-End Deep Learning
- Traditional + Learning

Bayesian approach

Inverse Problem Example: $\mathbf{y} = \mathbf{A}\mathbf{x}_g + \boldsymbol{\eta}$

$$\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \implies \quad \mathbf{y}|\mathbf{x} \sim \mathcal{N}(\mathbf{A}\mathbf{x}, \mathbf{I})$$

Bayesian approach

Inverse Problem Example: $\mathbf{y} = \mathbf{A}\mathbf{x}_g + \boldsymbol{\eta}$

$$\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \implies \quad \mathbf{y}|\mathbf{x} \sim \mathcal{N}(\mathbf{A}\mathbf{x}, \mathbf{I})$$

$$\mathbf{x}_{\text{MLE}} = \underset{\mathbf{x} \in \mathbb{R}^n}{\operatorname{argmax}} \underbrace{p(\mathbf{y}|\mathbf{x})}_{\text{Likelihood}} = \underset{\mathbf{x} \in \mathbb{R}^n}{\operatorname{argmin}} \underbrace{\frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2}_{\text{Data-fidelity}}$$

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In general, the data-fidelity term is denoted by $f(\mathbf{x}) = -\ln(p(\mathbf{y}|\mathbf{x}))$.

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In general, the data-fidelity term is denoted by $f(\mathbf{x}) = -\ln(p(\mathbf{y}|\mathbf{x}))$.

MAP estimate

- Choose a prior density $p(\mathbf{x})$ which promotes natural images.
- Set the function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ as $g(\mathbf{x}) = -\ln(p(\mathbf{x}))$.

$$\mathbf{x}_{\text{MAP}} = \underset{\mathbf{x} \in \mathbb{R}^n}{\operatorname{argmax}} \underbrace{p(\mathbf{y}|\mathbf{x})}_{\text{Likelihood}} \underbrace{p(\mathbf{x})}_{\text{Prior}} = \underset{\mathbf{x} \in \mathbb{R}^n}{\operatorname{argmin}} \underbrace{\frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2}_{\text{Data-fidelity}} + \underbrace{g(\mathbf{x})}_{\text{Regularizer}}$$

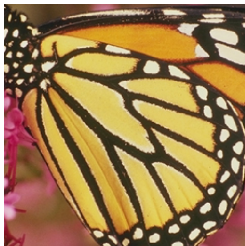
Optimization model

$$\operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^n} \underbrace{f(\mathbf{x})}_{\text{Data-fidelity}} + \underbrace{\lambda}_{\text{Regularization Parameter}} \underbrace{g(\mathbf{x})}_{\text{Regularizer}}$$

Optimization model

$$\operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^n} \underbrace{f(\mathbf{x})}_{\text{Data-fidelity}} + \underbrace{\lambda}_{\text{Regularization Parameter}} \underbrace{g(\mathbf{x})}_{\text{Regularizer}}$$

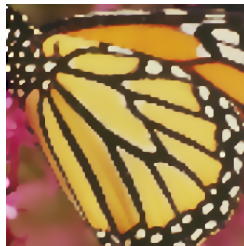
Total Variational Regularizer: $g(\mathbf{x}) = \sum_{i=1}^n \underbrace{|\nabla_h \mathbf{x}_i|}_{\text{Horizontal Gradient}} + \underbrace{|\nabla_v \mathbf{x}_i|}_{\text{Vertical Gradient}}.$



(a) Ground-Truth.



(b) MLE (PSNR 19dB).



(c) TV (PSNR 22dB).

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k),$$

$$\mathbf{x}_{k+1} = \operatorname{argmin}_{\mathbf{x}} \lambda \rho \mathbf{g}(\mathbf{x}) + \frac{1}{2} \|\mathbf{x} - \mathbf{z}_{k+1}\|^2.$$

BBS solver

$$\mathbf{z}_{k+1} = \operatorname{argmin}_{\mathbf{z}} f(\mathbf{z}) + \frac{\rho}{2} \|\mathbf{z} - \mathbf{x}_{k+1}\|^2,$$

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ADMM solver

$$\mathbf{v}_{k+1} = \operatorname{argmin}_{\mathbf{v}} f(\mathbf{v}) + \frac{\rho}{2} \|\mathbf{v} - (\mathbf{x}_k - \mathbf{u}_k)\|^2,$$

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$$\mathbf{u}_{k+1} = \mathbf{u}_k + (\mathbf{v}_{k+1} - \mathbf{x}_{k+1}).$$

Optimization model

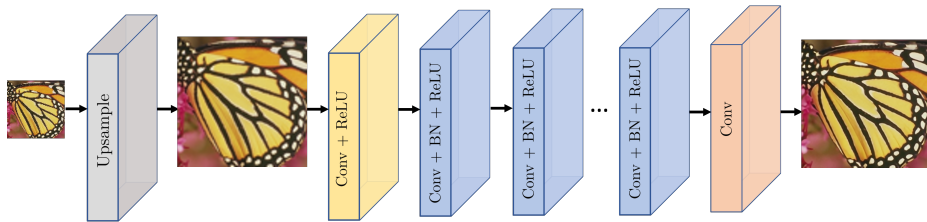
Advantages:

- Clear interpretation as inversion step is followed by regularization.
- Convergence of FBS, BBS and ADMM are well established.
- Flexibility in handling various image restoration tasks by simply specifying $\mathcal{H}$, noise model η and degraded measurements $\mathbf{y}$.

Disadvantages:

- Designing hand-crafted regularizers g is a difficult task.
- The parameters for the algorithms need to be tuned.
- Inference using sophisticated regularizers is time consuming.

Deep learning methods

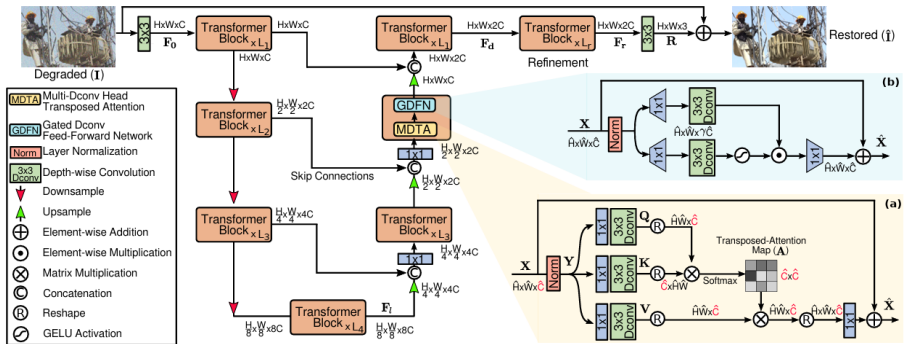


End-to-end CNN based network for superresolution.

Advantages:

- Good reconstruction performance.
- Fast inference time.

Deep learning methods



End-to-end attention based network for superresolution.

Advantages:

- State-of-the-art reconstruction performance.
- High inference time.

Disadvantage: Not robust to changes in forward model



(a) Ground-Truth.



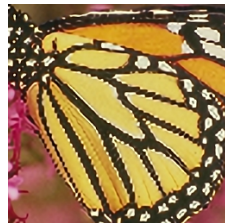
(b) **S**-bicubic (22dB).



(c) VDSR output (28dB).



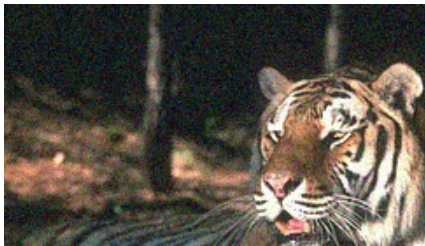
(d) **S**-decimation (19dB).



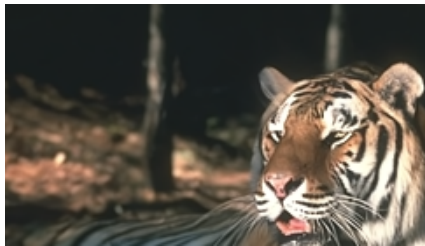
(e) VDSR output (16dB).

Measurements are obtained as $\mathbf{y} = \mathbf{S}\mathbf{x}$, where $\mathbf{x}$ is ground-truth image.

Disadvantage: Sensitive to minute perturbations in the input



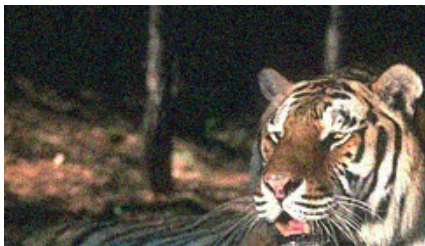
(a) Noisy input.



(b) Restormer output. PSNR: 34.2 dB.

Measurements: $\mathbf{y} = \mathbf{x} + \eta$ (Gaussian noise). Slight JPEG compression added to the input.

Disadvantage: Sensitive to minute perturbations in the input



(a) Noisy input.



(b) Restormer output. PSNR: 34.2 dB.



(c) Perturbed input (slight JPEG).



(d) Restormer output. PSNR: 27.9 dB.

Measurements: $\mathbf{y} = \mathbf{x} + \eta$ (Gaussian noise). Slight JPEG compression added to the input.

Traditional + Deep Learning

Optimization algorithms

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k),$$

$$\mathbf{x}_{k+1} = \underset{\mathbf{x}}{\operatorname{argmin}} \quad \lambda \rho \mathbf{g}(\mathbf{x}) + \frac{1}{2} \|\mathbf{x} - \mathbf{z}_{k+1}\|^2.$$

BBS solver

$$\mathbf{z}_{k+1} = \underset{\mathbf{z}}{\operatorname{argmin}} \quad f(\mathbf{z}) + \frac{\rho}{2} \|\mathbf{z} - \mathbf{x}_{k+1}\|^2,$$

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ADMM solver

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$$\mathbf{u}_{k+1} = \mathbf{u}_k + (\mathbf{v}_{k+1} - \mathbf{x}_{k+1}).$$

Definition

Given $g : \mathbb{R}^n \rightarrow \mathbb{R}$, proximal operator is defined as,

$$\text{prox}_g(\mathbf{y}) = \underset{\mathbf{x} \in \mathbb{R}^n}{\text{argmin}} \quad g(\mathbf{x}) + \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2.$$

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$$\mathbf{z}_{k+1} = \mathbf{x}_g + \boldsymbol{\eta}$$

$$\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \lambda \rho \mathbf{I})$$

$$p(\mathbf{x}) = \exp(-g(\mathbf{x}))$$

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$$\mathbf{x}^{k+1} = \mathbf{D}_{\lambda \rho}(\mathbf{z}^{k+1}), \text{ where } \mathbf{D} \text{ is a denoiser.}$$

$$\begin{aligned} \mathbf{z}_{k+1} &= \mathbf{x}_g + \boldsymbol{\eta} \\ \boldsymbol{\eta} &\sim \mathcal{N}(\mathbf{0}, \lambda \rho \mathbf{I}) \\ p(\mathbf{x}) &= \exp(-g(\mathbf{x})) \end{aligned}$$

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Replacing proximal operator by denoising

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FBS solver

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BBS solver

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ADMM solver

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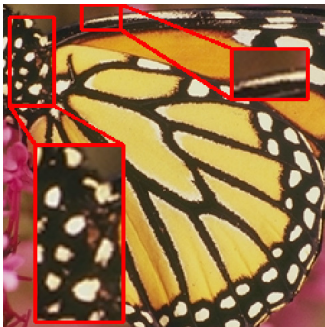
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Powerful current denoisers

1. Restormer [Zamir et al., CVPR 2022](#)
2. NAFNet [Chen et al., ECCV 2022](#)
3. PromptIR [Potlapalli et al., NeurIPS 2023](#)
4. MambaIR (State-space restoration), 2024
5. DRUNet [Zhang et al., IEEE TPAMI 2022](#)
6. DnCNN [Zhang et al., IEEE TIP 2017](#)
7. BM3D [Dabov et al., IEEE TIP 2007](#)

Use the best denoiser and are we done?



(a) Ground-Truth.

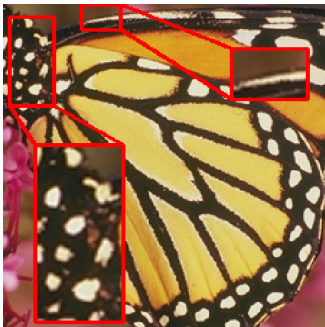


(b) TV (PSNR: 24dB).



(c) CNN (PSNR: 27dB).

Use the best denoiser and are we done?



(a) Ground-Truth.



(b) TV (PSNR: 24dB).



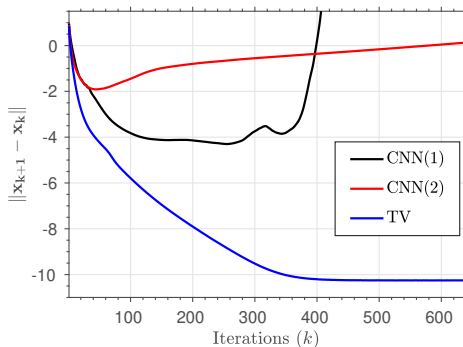
(c) CNN (PSNR: 27dB).

What if the iterations do not converge?

Why fuss about convergence?

Denoiser		Number of iterations				
		10	25	50	75	100
DnCNN	$\ \mathbf{x}_k - \mathbf{x}_{k-1}\ _2$	146.00	146.58	234.55	2300	2207248
	PSNR	16.54	14.79	-4.09	-45.14	-84.68
TV	$\ \mathbf{x}_k - \mathbf{x}_{k-1}\ _2$	74.06	30.65	7.67	1.99	0.52
	PSNR	20.42	22.96	23.69	23.74	23.74

Values of $\|\mathbf{x}_k - \mathbf{x}_{k-1}\|_2$ and PSNR (dB)

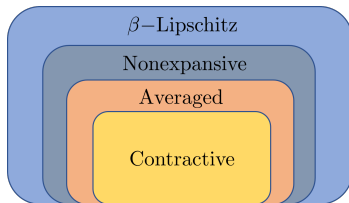


Key definitions

Special operators

An operator $\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be,

- β -Lipschitz where $\beta > 0$ if $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \|\mathbf{T}(\mathbf{x}) - \mathbf{T}(\mathbf{y})\| \leq \beta \|\mathbf{x} - \mathbf{y}\|$.
- nonexpansive if $\beta = 1$.
- contractive if $\beta \in [0, 1)$.
- θ -averaged for some $\theta \in (0, 1)$, if $\mathbf{T} = (1 - \theta)\mathbf{I} + \theta\mathbf{T}_n$ and $\mathbf{T}_n$ is nonexpansive.



Hierarchy of operators

Optimization Problem

$$\operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

Examples of contractive/averaged operators

Optimization Problem

$$\operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

Gradient descent iteration

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k) \quad \Rightarrow \quad \mathbf{x}_{k+1} = T_{\text{GD}}(\mathbf{x}_k), \text{ where } T_{\text{GD}} = \mathbf{I} - \rho \nabla f.$$

Examples of contractive/averaged operators

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Properties of T_{GD}

If f is convex with β -Lipschitz gradient, then for $\rho \in (0, 2/\beta)$, T_{GD} is,

- contractive when f is strongly convex.
- averaged when f is convex.

Examples of contractive/averaged operators

Optimization Problem

$$\operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) + g(\mathbf{x})$$

Examples of contractive/averaged operators

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FBS iteration

$$\begin{aligned} \mathbf{z}_{k+1} &= \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k), \\ \mathbf{x}_{k+1} &= \operatorname{prox}_{\lambda \rho g}(\mathbf{z}_{k+1}) \end{aligned} \quad \Rightarrow \quad \begin{aligned} \mathbf{x}_{k+1} &= T_{\text{FBS}}(\mathbf{x}_k), \text{ where} \\ T_{\text{FBS}} &= \operatorname{prox}_{\lambda \rho g} \circ (\mathbf{I} - \rho \nabla f). \end{aligned}$$

Examples of contractive/averaged operators

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If f is convex with β -Lipschitz gradient and g is convex, then, for $\rho \in (0, 2/\beta)$, T_{FBS} is,

- contractive when f is strongly convex.
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Key results

Banach contraction principle

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be contractive and let $\mathbf{x}_0 \in \mathbb{R}^n$. Then the sequence $(\mathbf{x}_k)_{k \geq 0}$ generated as $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$ converges to some $\mathbf{x}^*$ such that $T(\mathbf{x}^*) = \mathbf{x}^*$.

Key results

Banach contraction principle

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be contractive and let $\mathbf{x}_0 \in \mathbb{R}^n$. Then the sequence $(\mathbf{x}_k)_{k \geq 0}$ generated as $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$ converges to some $\mathbf{x}^*$ such that $T(\mathbf{x}^*) = \mathbf{x}^*$.

Proposed Lemma

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be θ -averaged and let $\mathbf{x}_0 \in \mathbb{R}^n$. Then the sequence $(\mathbf{x}_k)_{k \geq 0}$ generated as $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$ converges to some $\mathbf{x}^*$ such that $T(\mathbf{x}^*) = \mathbf{x}^*$ provided such a point exists.

Key results for PnP convergence

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k),$$

$$\mathbf{x}_{k+1} = \mathbf{D}(\mathbf{z}_{k+1})$$

BBS solver

$$\mathbf{z}_{k+1} = \text{prox}_{\frac{1}{\rho}f}(\mathbf{x}_k),$$

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Key contribution for PnP convergence

FBS update

$$\mathbf{x}_{k+1} = T_{\text{FBS}}(\mathbf{x}_k) = \mathbf{D}(\mathbf{I} - \rho \nabla f)(\mathbf{x}_k)$$

BBS update

$$\mathbf{x}_{k+1} = T_{\text{BBS}}(\mathbf{x}_k) = \mathbf{D}(\text{prox}_{\frac{1}{\rho}f})(\mathbf{x}_k)$$

Key contribution for PnP convergence

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Definition

Given an operator T , the set $\text{Fix}(T)$ consists of all points satisfying $\mathbf{x} = T(\mathbf{x})$.

Convergence Result 1

If $\mathbf{D}$ is contractive and f is convex

- FBS update $\mathbf{x}_k$ converges to a unique $\mathbf{x}^* \in \text{Fix}(T_{\text{FBS}})$
- BBS update $\mathbf{x}_k$ converges to a unique $\mathbf{x}^* \in \text{Fix}(T_{\text{BBS}})$

Convergence guarantees with existing denoisers

Optimality

- Sreehari et al. (TCI '16).
- Teodoro et al. (TIP '19).
- Hurault et al. (ICLR '22).

First two works restricted to symmetric denoisers.

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Strong/unverifiable assumptions.

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Strong/unverifiable assumptions.

Optimality and iterative convergence

- Pravin et al. (TCI '21, TIP '22)

Restricted to kernel filters

Why learning-based denoisers are required?

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In cases where iterations converge, learning based denoisers give much better performance than kernel denoisers.



(a) Ground Truth $\mathbf{x}$. (b) $\mathbf{y}$ (c) Upsampled input. (d) NLM(PSNR 22.81dB). (e) CNN(PSNR 24.88dB).

Superresolution with downsampling factor 4 and additive Gaussian noise

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Superresolution with downsampling factor 4 and additive Gaussian noise

	NLM	CNN
PSNR	23.37	25.30
SSIM	0.6733	0.7529

Challenges of training contractive/averaged CNN denoisers

- 1) Computing Lipschitz for CNN is not tractable.
- 2) There is clear trade-off between denoising accuracy and CNN denoiser being nonexpansive or 1-Lipschitz (decrease in denoising accuracy upto 7-8 dB).¹
- 3) Training CNN to be nonexpansive is a challenging task.²
- 4) Existing algorithms to force CNN to be nonexpansive either fail to provide guarantees or are computationally very expensive

¹Hurault et al., "Proximal denoiser for convergent plug-and-play optimization with nonconvex regularization.", arXiv preprint arXiv:2201.13256 (2022)

²Hertrich et al., "Convolutional proximal neural networks and plug-and-play algorithms.", Linear Algebra Appl., 631, 203234 (2021).

Previous works for constructing nonexpansive denoisers

No guarantees

- Ryu et al. (ICML '19).
- Pesquet et al. (SIAM '21).
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Training is computationally expensive

- Hertrich, et al. (LAA '21).
- Terris et al. (ICASSP '20).
- Sedghi et al. (ICLR '19).

Goals of this work

- Construct learning-based denoisers which are provably contractive and provide near to best denoising performance.
- The final iterative algorithms should give state-of-the-art performance for image reconstruction.

Gaussian denoising: $\mathbf{y} = \mathbf{x} + \eta$

Data-fidelity term: $\ell(\mathbf{x}; \mathbf{y}) = \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|^2$.

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Consider wavelet based regularizer.

$$g(\mathbf{x}) = \|\mathbf{W}\mathbf{x}\|_{\Lambda} := \sum_{j=1}^p \lambda_j |(\mathbf{W}\mathbf{x})_j|,$$

where,

- $\Lambda \in \mathbb{R}^{p \times p}$ is a diagonal positive definite matrix such that $\Lambda_{jj} := \lambda_j$.
- $\mathbf{W} \in \mathbb{R}^{p \times p}$ is chosen to be an orthogonal wavelet transform.

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Optimization problem: $\underset{\mathbf{x}}{\operatorname{argmin}} \ell(\mathbf{x}; \mathbf{y}) + g(\mathbf{x})$

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla \ell(\mathbf{x}_k; \mathbf{y}),$$

$$\mathbf{x}_{k+1} = \text{Prox}_g(\mathbf{z}_{k+1}).$$

Applying FBS for denoising

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla \ell(\mathbf{x}_k; \mathbf{y}),$$

$$\mathbf{x}_{k+1} = \text{Prox}_g(\mathbf{z}_{k+1}).$$

$$\mathbf{x} - \rho \nabla \ell(\mathbf{x}; \mathbf{y}) = (1 - \rho)\mathbf{x} + \rho \mathbf{y}$$

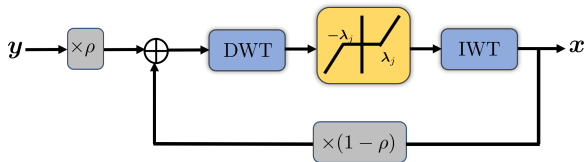
$$\text{Prox}_g(\mathbf{z}) = \underset{\mathbf{x}}{\text{argmin}} \frac{1}{2} \|\mathbf{z} - \mathbf{x}\|^2 + \|\mathbf{W}\mathbf{x}\|_{\Lambda} = \underbrace{\mathbf{W}^{\top} \mathbf{S}_{\Lambda}(\mathbf{W}\mathbf{z})}_{\text{Wavelet Denoising}}$$

- $\mathbf{S}_{\Lambda}(\mathbf{x})$ consists of $\mathbf{S}_{\lambda_j}(x_j)$ with $\mathbf{S}_{\lambda_j}$ as soft thresholding operations of $\mathbf{x}_j$ with threshold λ_j and $\mathbf{x}_j$ is j th element of $\mathbf{x}$.

$$\mathbf{z}_{k+1} = (1 - \rho)\mathbf{x}_k + \rho\mathbf{y},$$

$$\mathbf{x}_{k+1} = \mathbf{W}^\top \mathbf{S}_\Lambda(\mathbf{W}\mathbf{z}_{k+1})$$

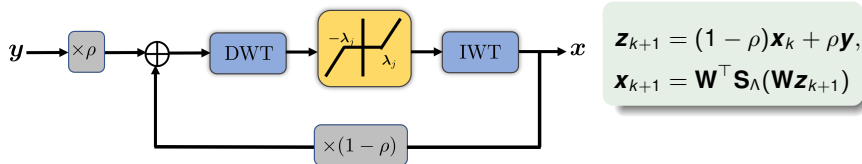
FBS network



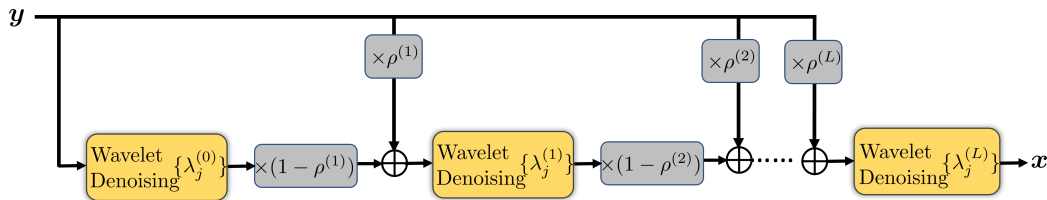
$$\mathbf{z}_{k+1} = (1 - \rho)\mathbf{x}_k + \rho\mathbf{y},$$
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FBS algorithm for Gaussian denoising

FBS network



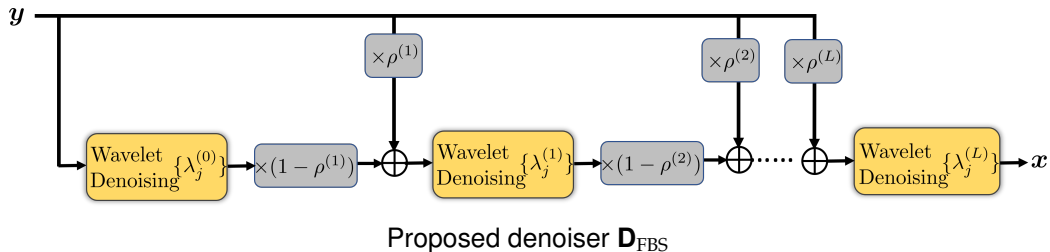
FBS algorithm for Gaussian denoising



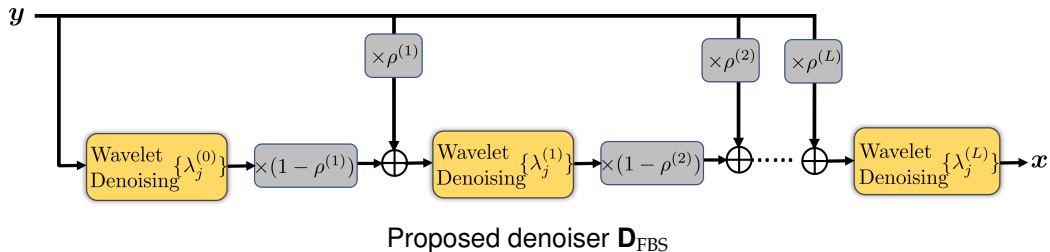
Unfolding FBS algorithm for Gaussian denoising

* Lecun et al. used unfolding ISTA algorithm and learn parameters for compressed sensing in "Learning fast approximations of sparse coding.", Proc. Intl. Conf. Mach. Learn., 2010.

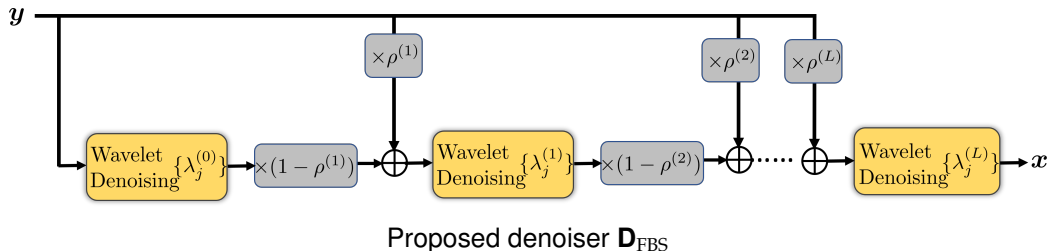
Deep unfolding network



Deep unfolding network



Deep unfolding network

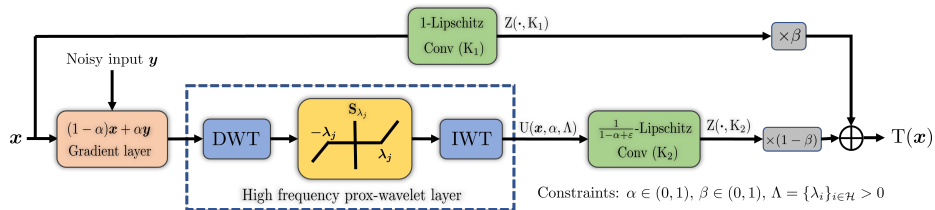


Proposition

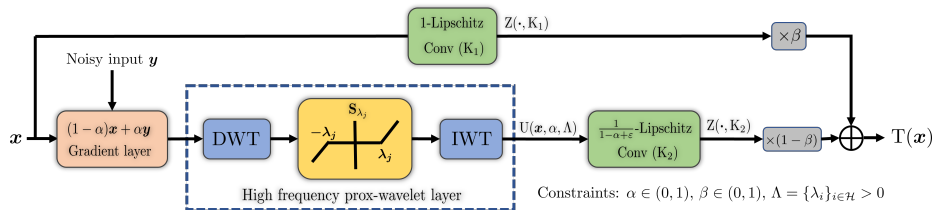
Let $\Lambda^{(\ell)}$ be diagonal positive definite matrix, $\mathbf{W}^{(\ell)}$ be any orthogonal wavelet transform and $\rho^{(\ell)} \in (0, 1)$ for all $\ell \in \{1, 2, \dots, L\}$, then the denoiser is contractive, and hence averaged and nonexpansive.

- Denoising accuracy is off by only 1.5 – 4 dB compared to the best denoisers.
- Convergence of atleast 5 reconstruction algorithms including PnP-ISTA, RED-PG!!!

Refinements to network reduces denoising gap to 1 – 2 dB

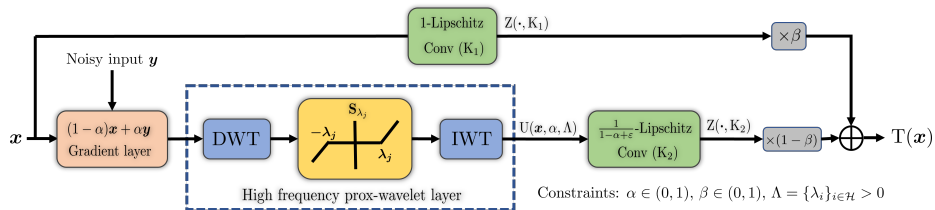


Refinements to network reduces denoising gap to 1 – 2 dB



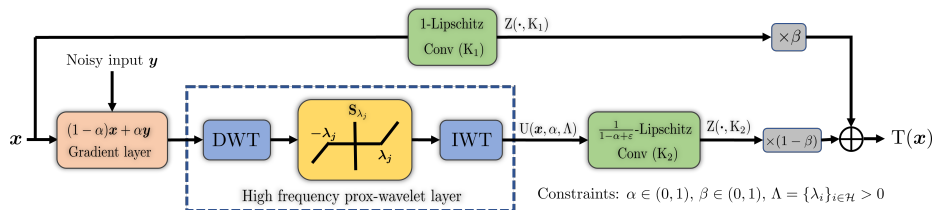
- **Operator norm controls Lipschitzness:** For convolution $\mathbf{V}_K$, let $s(K) := \|\mathbf{V}_K\|_{\text{op}}$. The layer is 1-Lipschitz (nonexpansive) iff $s(K) \leq 1$.

Refinements to network reduces denoising gap to 1 – 2 dB



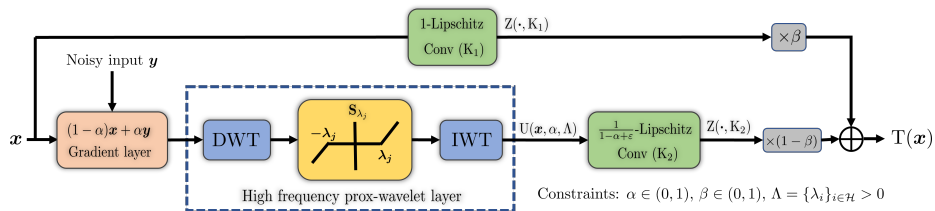
- **Operator norm controls Lipschitzness:** For convolution $\mathbf{V}_K$, let $s(K) := \|\mathbf{V}_K\|_{\text{op}}$. The layer is 1-Lipschitz (nonexpansive) iff $s(K) \leq 1$.
- **Normalize the convolution:** Define $\bar{\mathbf{V}}_K := \mathbf{V}_K / s(K)$, so $\|\bar{\mathbf{V}}_K\|_{\text{op}} \leq 1$ and $\mathbf{Z}(x; K) = \bar{\mathbf{V}}_K x = \mathbf{V}_K x / s(K)$.

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- Make it contractive / set a budget:** Use $\frac{1}{s(K) + \epsilon}$ with $\epsilon > 0$ for a strict contraction, and scale by γ to obtain a γ -Lipschitz layer: $\gamma \mathbf{Z}(x; K)$.

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H. Sedghi, V. Gupta, and P. M. Long, **The Singular Values of Convolutional Layers**, *Proc. International Conference on Learning Representations*, 2019.

Theoretical results

FBS solver

$$\mathbf{z}_{k+1} = \mathbf{x}_k - \rho \nabla f(\mathbf{x}_k),$$

$$\mathbf{x}_{k+1} = \mathbf{D}_{\text{FBS}}(\mathbf{z}_{k+1}),$$

BBS solver

$$\mathbf{z}_{k+1} = \text{prox}_{\frac{1}{\rho}f}(\mathbf{x}_k),$$

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BBS solver

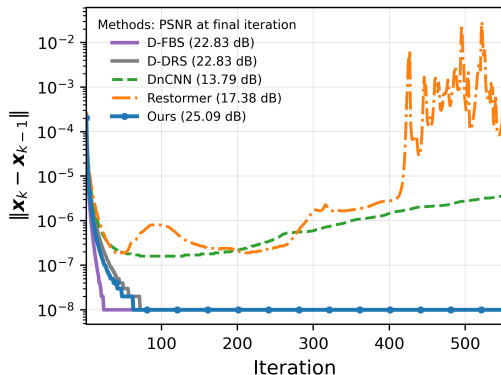
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$$\mathbf{x}_{k+1} = \mathbf{D}_{\text{FBS}}(\mathbf{z}_{k+1}),$$

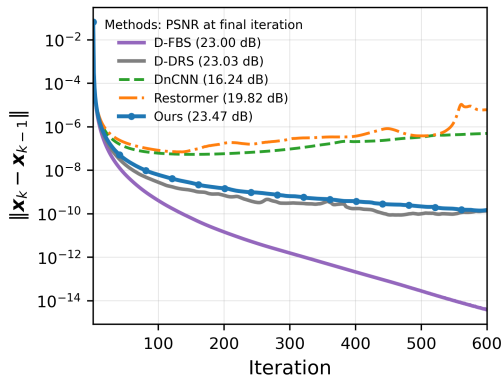
Convergence Results

- If f is convex and differentiable with β -Lipschitz gradient, FBS update $\mathbf{x}_k$ converges to a unique $\mathbf{x}^* \in \text{Fix}(\mathcal{T}_{\text{FBS}})$
- If f is convex, BBS update $\mathbf{x}_k$ converges to a unique $\mathbf{x}^* \in \text{Fix}(\mathcal{T}_{\text{BBS}})$

Validating theoretical results



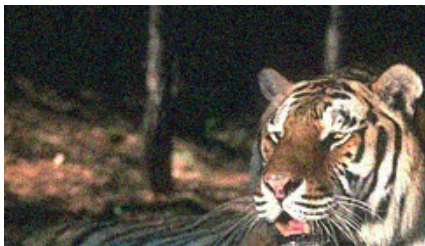
(a) Superresolution



(b) Deblurring

Convergence within PnP-FBS. 1-Lipschitz denoisers converge, with ours achieving the highest PSNR asymptotically. Unconstrained models like DnCNN and Restormer diverge.

Disadvantage: Sensitive to minute perturbations in the input



Noisy input.



Restormer output. PSNR: 34.2 dB.



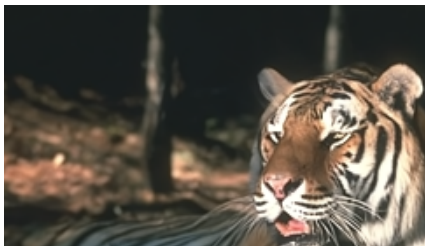
Perturbed input (slight JPEG).



Restormer output. PSNR: 27.9 dB.

Measurements: $\mathbf{y} = \mathbf{x} + \boldsymbol{\eta}$ (Gaussian noise). Slight JPEG compression added to the input.

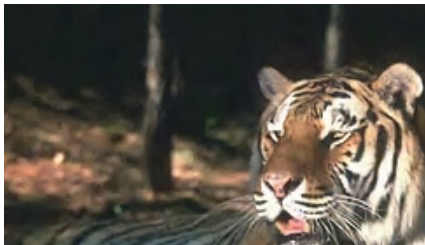
Our model is not sensitive to minute perturbations in the input



Restormer output before. PSNR: 34.2 dB.



Restormer output after. PSNR: 27.9 dB.



Our output before. PSNR: 33.2 dB.



Our output after. PSNR: 32.2 dB.

Measurements: $\mathbf{y} = \mathbf{x} + \eta$ (Gaussian noise). Slight JPEG compression added to the input.

Color denoising: PSNR(dB) / SSIM

Dataset, σ	Methods		
	Ours	DnCNN	Restormer
CBSD68, $\sigma=15$	32.85/0.9158	33.63/0.9296	33.85/0.9316
CBSD68, $\sigma=25$	29.83/0.8549	30.72/0.8806	30.90/0.8837
CBSD68, $\sigma=50$	25.61/0.7211	26.52/0.7714	26.65/0.7765
Kodak24, $\sigma=15$	33.52/0.9066	34.37/0.9219	34.62/0.9237
Kodak24, $\sigma=25$	30.72/0.8495	31.70/0.8837	31.93/0.8814
Kodak24, $\sigma=50$	26.49/0.7285	27.63/0.7785	27.82/0.7901
Urban100, $\sigma=15$	31.96/0.9092	32.65/0.9199	33.16/0.9229
Urban100, $\sigma=25$	28.94/0.8585	30.11/0.8837	30.55/0.8900
Urban100, $\sigma=50$	24.40/0.7441	25.74/0.7990	26.07/0.8129
DIV2K, $\sigma=15$	33.71/0.9020	34.40/0.9131	34.62/0.9144
DIV2K, $\sigma=25$	30.79/0.8511	31.67/0.8717	31.87/0.8751
DIV2K, $\sigma=50$	26.10/0.7430	26.95/0.7828	27.09/0.7889
McMaster, $\sigma=15$	32.34/0.8774	33.08/0.8895	33.51/0.8944
McMaster, $\sigma=25$	29.45/0.8093	30.54/0.8364	31.82/0.8414
McMaster, $\sigma=50$	24.82/0.6761	25.92/0.7216	26.11/0.7271

Empirical Results: Deblurring



Blurred (16.55 dB / 0.3440)



Restormer (25.11 dB / 0.6901)



DnCNN (25.36 dB / 0.7504)



Ours (25.16 dB / 0.7515)

Empirical Results: $2\times$ Superresolution



Bicubic (23.79 / 0.696)



DnCNN (26.78 / 0.759)



Restormer (25.23 / 0.732)



Ours (26.84 / 0.783)

Conclusions

- Convergence analysis of PnP algorithms is vital since iterative techniques which do not converge produces spurious reconstructions.
- There exists trade-off between denoising performance and nonexpansivity of learning based denoisers.
- Constructed contractive Gaussian denoisers for regularization with standalone competitive denoising performance.
- Contractive denoisers are special because of unique and linear convergence of PnP algorithms.
- Algorithms developed are both convergent and competitive with state-of-the-art performance for image reconstruction.

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Acknowledgments

First half of the work

Prof. Kunal N. Chaudhury, *Department of Electrical Engineering, IISc, Bengaluru.*

- P. Nair, R. G. Gavaskar, and K. N. Chaudhury, “Fixed-point and objective convergence of plug-and-play algorithms,” *IEEE Transactions on Computational Imaging*, 2021.
- P. Nair and K. N. Chaudhury, “Averaged deep denoisers for image regularization,” *Journal of Mathematical Imaging and Vision*, 66(3):362–379, 2024.

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Second half of the work

Shubhi Shukla, *Algorithms for Imaging Sciences Laboratory, IIT Madras.*

- S. Shukla and P. Nair, “Provably Contractive and High-Quality Denoisers for Convergent Restoration,” *Under review at CVPR (2026)*.

Thank You!!